SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA1704

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4   Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **R S Jahnavi (192371056)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:R S JAHNAVI

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1. P1: SoilDry → IrrigationNeeded
2. P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3. P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4. Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1. Stench [1,2] → Wumpus is adjacent to cell [1,2]
2. Breeze [1,1] → Pit is adjacent to cell [1,1]
3. Glitter [x,y] → Gold is at [x,y]
4. ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — SMART MEDICAL DIAGNOSIS SYSTEM

1.1 Problem Understanding

The hospital uses a rule-based Logical Agent to provide preliminary diagnoses from patient symptoms. The knowledge base contains rules connecting symptoms such as fever, cough, rash, and breathlessness with possible diseases. Patient A is known to have fever, cough, and breathlessness.

1.2 (a) Propositional Logic Representation

Definition of Symbols

Let **F** represent "Patient A has Fever," **C** represent "Patient A has Cough," **B** represent "Patient A has Breathlessness," **R** represent "Patient A has Rash," **Flu** represent "Patient A has Possible Flu," **M** represent "Patient A has Possible Measles," and **P** represent "Patient A has Possible Pneumonia."

Logical Rules

The given rules can be represented as follows:

- **Rule 1:** $F \land C \rightarrow Flu$
- **Rule 2:** $F \land R \rightarrow M$
- **Rule 3:** $C \land B \rightarrow P$

Given Facts

- F = True
- C = True
- B = True

Here, $\wedge$ means AND and $\rightarrow$ means implication.

1.3 (b) Forward Chaining

Initial Facts

Patient A has Fever, Cough, and Breathlessness.

Inference 1 — Rule 1

Since **F = True** and **C = True**, Rule 1 is satisfied:

F $\wedge$ C $\rightarrow$ Flu

Therefore:

Flu = True

Inference 2 — Rule 3

Since **C = True** and **B = True**, Rule 3 is satisfied:

C $\wedge$ B $\rightarrow$ Pneumonia

Therefore:

Pneumonia = True

Inference 3 — Rule 2

Rule 2 requires both Fever and Rash:

F $\wedge$ R $\rightarrow$ Measles

Although Fever is true, Rash is not given. Therefore, **Measles cannot be derived**.

Result

The system derives **Possible Flu** and **Possible Pneumonia**, but it cannot derive Possible Measles.

1.4 (c) Backward Chaining

Goal

The goal is:

Pneumonia(A)

The rule that produces Pneumonia is:

Cough(A) $\wedge$ Breathlessness(A) $\rightarrow$ Pneumonia(A)

Therefore, the goal is reduced into two sub-goals: **Cough(A)** and **Breathlessness(A)**. Both are already known facts. Hence, the goal is successfully established.

Goal Reduction Tree

```
          Pneumonia(A)
               |
      Cough(A) ∧ Breathlessness(A)
          /           \
```

```
       /             \
  Cough(A)        Breathlessness(A)
    TRUE              TRUE
       \            /
        \          /
       Pneumonia(A)
          TRUE
```

Conclusion

Therefore, **Patient A has Possible Pneumonia** according to the given knowledge base.

CASE STUDY 2 — AUTONOMOUS TRAFFIC MANAGEMENT AGENT

2.1 Problem Understanding

The smart city traffic controller uses First-Order Logic to manage emergency vehicle movement. The knowledge base contains rules for identifying emergency vehicles, clearing their paths, providing green signals, and allowing vehicles behind them to proceed.

2.2 (a) Unification

Rule 1

∀x [Vehicle(x) ∧ Emergency Type(x) → ClearPath(x)]

Matching with Facts

Given:

- Vehicle(Ambulance)
- EmergencyType(Ambulance)

The variable **x** is matched with **Ambulance**.

Therefore:

θ = {x/Ambulance}

Substitution Steps

Step	Rule Expression	Fact	Substitution
1	Vehicle(x)	Vehicle(Ambulance)	x/Ambulance
2	EmergencyType(x)	EmergencyType(Ambulance)	x/Ambulance
3	ClearPath(x)	—	ClearPath(Ambulance)

Thus, the MGU is:

θ = {x/Ambulance}

and we derive:

ClearPath(Ambulance)

2.3 (b) Resolution Method

CNF Conversion

Rule 1 becomes:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2 becomes:

¬ClearPath(x) ∨ GreenSignal(x)

and

¬ClearPath(x) ∨ ¬RedSignal(x)

Rule 3 becomes:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Given Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

Negated Goal

Goal:

Proceed(CarA)

Negated goal:

¬Proceed(CarA)

Resolution Steps

From Rule 1 and the ambulance facts, we derive:

ClearPath(Ambulance)

Using Rule 2:

ClearPath(Ambulance) → GreenSignal(Ambulance)

Therefore:

GreenSignal(Ambulance)

Now Rule 3 can be applied because:

- Vehicle(CarA) is true.
- Behind(CarA, Ambulance) is true.
- GreenSignal(Ambulance) is true.

Therefore:

Proceed(CarA)

Finally:

Proceed(CarA) + ¬Proceed(CarA) → □

Conclusion

The empty clause □ is obtained, proving that:

Proceed(CarA) is true.

2.4 (c) Two Simultaneous Emergency Vehicles

Analysis

The current knowledge base can identify multiple emergency vehicles independently. For example, if Ambulance1 and Ambulance2 are both emergency vehicles, Rule 1 can derive a clear path for both.

However, the knowledge base does not contain rules for situations where the two emergency vehicles have conflicting routes or reach the same intersection simultaneously.

Additional Requirements

The system should include:

- Emergency vehicle priority rules.
- Route conflict detection.
- Collision avoidance rules.
- Vehicle location and direction facts.
- Signal coordination rules.

Conclusion

Therefore, the current knowledge base is **partially sufficient**, but additional rules are needed for safe coordination of simultaneous emergency vehicles.

CASE STUDY 3 — AGRICULTURAL EXPERT REASONING SYSTEM

3.1 Problem Understanding

The agricultural advisory system uses logical rules to determine whether irrigation is required and whether the drip irrigation method should be applied. The given facts indicate that the soil is dry and the crop is wheat.

3.2 (a) CNF Conversion

Rule P1

Original:

SoilDry → IrrigationNeeded

Eliminating implication:

¬SoilDry ∨ IrrigationNeeded

Therefore:

CNF(P1) = ¬SoilDry ∨ IrrigationNeeded

Rule P2

Original:

IrrigationNeeded ∧ CropWheat → ApplyDripMethod

After eliminating implication:

¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod

Using De Morgan's law:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

Therefore:

CNF(P2) = ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

Rule P3

Original:

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

After eliminating implication:

¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk

Moving negation inward:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Therefore:

CNF(P3) = ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Facts

The facts are:

SoilDry

and

CropWheat

Final CNF

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
4. SoilDry
5. CropWheat

3.3 (b) Resolution Refutation

Goal

The goal is:

ApplyDripMethod

Negated goal:

¬ApplyDripMethod

Resolution Step 1

Using:

¬SoilDry ∨ IrrigationNeeded

and:

SoilDry

we derive:

IrrigationNeeded

Resolution Step 2

Using:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

and:

IrrigationNeeded

we derive:

¬CropWheat ∨ ApplyDripMethod

Resolution Step 3

Using:

CropWheat

we derive:

ApplyDripMethod

Resolution Step 4

Resolve:

ApplyDripMethod

with:

¬ApplyDripMethod

Therefore:

□

Resolution Tree

```
        SoilDry
           |
           ↓
    IrrigationNeeded
           |
           ↓
       CropWheat
           |
           ↓
    ApplyDripMethod
           |
```

+
|
¬ApplyDripMethod
Negated Goal
|
↓
□

Conclusion

The empty clause is obtained, so **ApplyDripMethod is logically proven**.

3.4 (c) Removing SoilDry

Step 1

If **SoilDry** is removed, Rule P1 cannot be activated.

Therefore:

IrrigationNeeded cannot be derived.

Step 2

Since IrrigationNeeded is unavailable, Rule P2 cannot establish:

ApplyDripMethod

Step 3

The absence of ApplyDripMethod does not mean:

¬ApplyDripMethod

Therefore, Rule P3 cannot be applied.

Final Analysis

After removing SoilDry:

- IrrigationNeeded → **Not derivable**
- ApplyDripMethod → **Not derivable**
- ¬ApplyDripMethod → **Not given**
- CropAtRisk → **Not derivable**

Conclusion

Therefore, **CropAtRisk does not follow** from the remaining knowledge base.

CASE STUDY 4 — WUMPUS WORLD KNOWLEDGE AGENT

4.1 Problem Understanding

The Wumpus World agent receives environmental percepts and uses propositional logic to reason about dangerous locations and the position of gold. The three percepts are Stench at [1,2], Breeze at [1,1], and Glitter at [2,2].

4.2 (a) Belief State Construction

Initial Percepts

The agent knows:

- **Stench[1,2]**
- **Breeze[1,1]**
- **Glitter[2,2]**

Inference from Stench

Using:

Stench[1,2] → Wumpus is adjacent to [1,2]

the agent concludes that a Wumpus may be in one of the adjacent cells:

[1,1], [1,3], [2,2]

Inference from Breeze

Using:

Breeze[1,1] → Pit is adjacent to [1,1]

the possible Pit locations are:

[1,2], [2,1]

Inference from Glitter

Using:

Glitter[x,y] → Gold[x,y]

the agent concludes:

Gold[2,2]

Derived Knowledge

Therefore, the agent's belief state contains:

- Possible Wumpus locations: **[1,1], [1,3], [2,2]**
- Possible Pit locations: **[1,2], [2,1]**
- Gold location: **[2,2]**

4.3 (b) Forward Chaining

Step 1 — Stench

Stench[1,2]

↓

Wumpus adjacent to [1,2]

↓

Possible Wumpus:

[1,1], [1,3], [2,2]

Step 2 — Breeze

Breeze[1,1]

↓

Pit adjacent to [1,1]

↓

Possible Pit:

[1,2], [2,1]

Step 3 — Glitter

Glitter[2,2]

↓

Gold[2,2]

Thus, Forward Chaining produces the complete set of possible dangerous locations and confirms the gold location.

4.4 (c) Safety Analysis of the Path

Cell [1,1]

The Breeze at [1,1] indicates that a pit is adjacent to [1,1]. It does not directly prove that [1,1] contains a pit. Therefore, the cell is not proven dangerous.

Cell [1,2]

The cell [1,2] is a possible Pit location because of the Breeze at [1,1]. Therefore, the agent cannot prove that [1,2] is safe.

Cell [2,2]

The Glitter confirms that gold is present at [2,2]. However, [2,2] is also a possible Wumpus location based on the Stench at [1,2]. Therefore, the cell cannot be guaranteed safe.

Conclusion

The path:

[1,1] → [1,2] → [2,2]

is **not guaranteed to be safe** with the current knowledge. The agent should gather additional information before moving through [1,2] and [2,2].